TOPIC 5 · COPILOT AGENTS AND MINI CAPSTONE
LAB 12 THEME · MINI CAPSTONE

Lab 12 — Mini Capstone: Group Dashboard and Decision Deck

Maps to	LO5: build a grounded Copilot agent and apply the full workflow in a group capstone
Duration	45 minutes
Tools	Excel Copilot · PowerPoint Copilot · Microsoft 365 Copilot web · OneDrive
Mock data	Your own mock or sanitised dataset
Course	GenAI for Engineers · v2.3
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-12

Goal

Per group: a mock or sanitised data table, a visible .Rules sheet, an interactive dashboard (six KPI tiles, three charts, slicers), a five-slide decision deck and a three-minute showcase.

What You Will Do

Work in four groups. Each group brings its own mock or sanitised dataset, writes .Rules for it, uses a dashboard Skill to build an interactive Excel dashboard with KPI tiles, charts and slicers, then turns the verified evidence into a short PowerPoint decision deck and presents it in a three-minute showcase. Use no confidential or personal data.

What You Will Build

Per group: a mock or sanitised data table, a visible .Rules sheet, an interactive dashboard (six KPI tiles, three charts, slicers), a five-slide decision deck and a three-minute showcase.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Labs 4-10 methods: .Rules, Skills, interactive dashboard and decision deck.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. (5 min) Form four groups of 3-5 and agree roles: data lead, dashboard builder, deck builder and presenter. Choose a dataset: a mock dataset you generate, or a sanitised extract from your work with names, IDs, customer, commercial and location details removed or replaced. Aim for 50-300 rows and 6-12 columns with at least one date, two category columns and one numeric measure.

PROMPT / CHECK — copy and adapt

```
MICROSOFT 365 COPILOT WEB:
Create a realistic but fictional dataset for an engineering team in [your domain, for example
lift maintenance, site inspections or equipment downtime]. Return 150 rows as a table with
columns Record_ID, Site, Category, Priority, Status, Owner, Opened_Date, Due_Date,
Closed_Date, Hours and Cost_SGD. Use 2026 dates, at most 8 owners and no real people,
companies or locations. State that the data is synthetic.
```

Step 2. (3 min) In Excel paste or open the data, convert it to a table (Ctrl+T) named tblData, add a Control sheet with your reporting date in B2 and save Group<N>_Dashboard.xlsx in your OneDrive work-genai folder. Profile the data before building.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
Profile tblData: row count, duplicate IDs, blank values per column, category values and date
range. Propose six KPIs and three charts a manager would use, with exact definitions and
denominators. Do not edit.
```

Step 3. (4 min) Create a visible .Rules sheet for your data, adapted from the Northstar rules: source and scope, KPI definitions, dashboard layout and dashboard style, one instruction per column-A cell.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
Create a visible worksheet named exactly .Rules. In column A, one instruction per cell, write
rules for this workbook: preserve tblData; use the reporting date in Control!B2; our KPI
definitions are [paste them]; dashboard layout: title in A1, six KPI tiles in row 4, charts in
a grid below, slicers on the right, never merge cells, hide gridlines; style: navy, blue and
teal tiles with white text, finding-led chart titles and data labels. Do not change any data.
```

Step 4. (10 min) Install data/templates/interactive-dashboard/SKILL.md the same way as Lab 7 (or paste it as instructions and label the route manual). Plan, review, then build. Polish the charts with @chart-design.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@interactive-dashboard Plan a one-page interactive dashboard for tblData using the visible
.Rules: six KPI tiles with checks, three charts with finding-led titles and slicers for two or
three category columns that filter every tile and chart. Do not edit yet.

EXCEL - ALLOW EDITING:
@interactive-dashboard Apply the reviewed plan. Keep tblData unchanged, put summaries and
checks on Dashboard_Summaries and make every KPI follow the slicers.

EXCEL - ALLOW EDITING:
@chart-design Review every dashboard chart: finding-style titles, data labels and the .Rules
colours. Do not change any values.
```

Step 5. (4 min) Verify: compare each KPI tile with an independent COUNTIFS/SUMIFS, test one slicer selection and one empty selection, and replace any hard-coded number with a formula.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
Reconcile every KPI tile with an independent formula from tblData with all slicers cleared,
then for one slicer selection. List mismatches, hard-coded values and fixed ranges. Do not
edit.
```

Step 6. (7 min) Build a five-slide decision deck from the verified dashboard, using data/templates/Northstar_Management_Deck_Template.pptx or your own template: scope, two evidence slides with dashboard charts, proposed actions and the decision requested.

PROMPT / CHECK — copy and adapt

```
POWERPOINT COPILOT:
Using the verified dashboard values I paste below, create a five-slide decision deck: 1 title, scope and as-at date; 2-3 one finding per slide with the matching chart; 4 proposed actions labelled proposed; 5 decision requested. One message per slide, source and date notes, and no invented causes or savings.
```

Step 7. (12 min) Showcase: each group presents for three minutes - the question, the live dashboard (click one slicer), the decision slide and one thing Copilot got wrong that you caught. The other groups give one strength and one question. Afterwards remove any real data you are not authorised to keep.

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

Copy-ready prompts: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-12

#	Purpose	Prompt
1	Data	Create a fictional 150-row engineering dataset; no real people, companies or locations.
2	Dashboard	@interactive-dashboard Plan six KPI tiles, three charts and slicers for tblData.

Test It

Check	Expected result
Safe data	Mock or sanitised only; no personal, customer or confidential data.
Dashboard	KPI tiles reconcile to independent formulas; charts and slicers filter together.
Deck	Every number matches the dashboard; actions labelled proposed; decision requested.
Showcase	Three-minute demo with one live slicer interaction and one caught Copilot error.

If You Get Stuck

Short of time? Present the Lab 8 Northstar dashboard and your Lab 10 deck instead, and say so in the showcase.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Turn your group's .Rules and dashboard Skill into a reusable pair your team can apply to next month's data.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?